

Research Proposal: Reliability of Information Criteria in Determining VAR Lag-Orders for Macroeconomic IRF Analysis

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Abstract

Briefly summarize the motivation, methodology, and expected outcomes of your Monte Carlo simulation project testing VAR lag-order information criteria (AIC, SIC, HQC) and their effect on Impulse Response Functions (IRFs) or Persistence Profiles.

1 Introduction

Introduce the importance of Vector Autoregression (VAR) and VECM models in macroeconomics (e.g., Johansen-Juselius 1990, Ramey 2011). Discuss the challenge of selecting the correct lag order and how incorrect choices impact structural analysis and policy implications.

2 Literature Review

Review existing literature on lag selection criteria. Discuss the theoretical trade-offs between the Akaike Information Criterion (AIC), Schwarz/Bayesian Information Criterion (SIC/BIC), and Hannan-Quinn Criterion (HQC) in small versus large samples.

3 Methodology

3.1 Data Generating Processes (DGPs)

Describe how you will construct your DGPs based on empirical macroeconomic datasets (e.g., extracting parameters by force-fitting models on the Johansen-Juselius dataset and

Ramey 2011 dataset).

3.2 Monte Carlo Simulation Design

Explain the core simulation loop:

- Drawing synthetic Gaussian innovations based on empirical covariance matrices.
- Generating synthetic time series of length T .
- Fitting restricted VAR/VECM models for test lags 1 to p_{max} .
- Recording the lag selected by AIC, SIC, and HQC.
- Calculating the Mean Squared Error (MSE) of the resulting IRFs/Persistence Profiles compared to the true profile.

4 Expected Results

Discuss your hypotheses. For example, do you expect AIC to over-parameterize the model? Do you expect SIC to be more accurate in large samples but underfit in small samples ($T = 80$)? How will these selection errors specifically distort the estimated Impulse Response Functions?

5 Timeline

Provide a brief schedule for completing the DGP extractions, Monte Carlo simulation coding in R, and the final write-up.